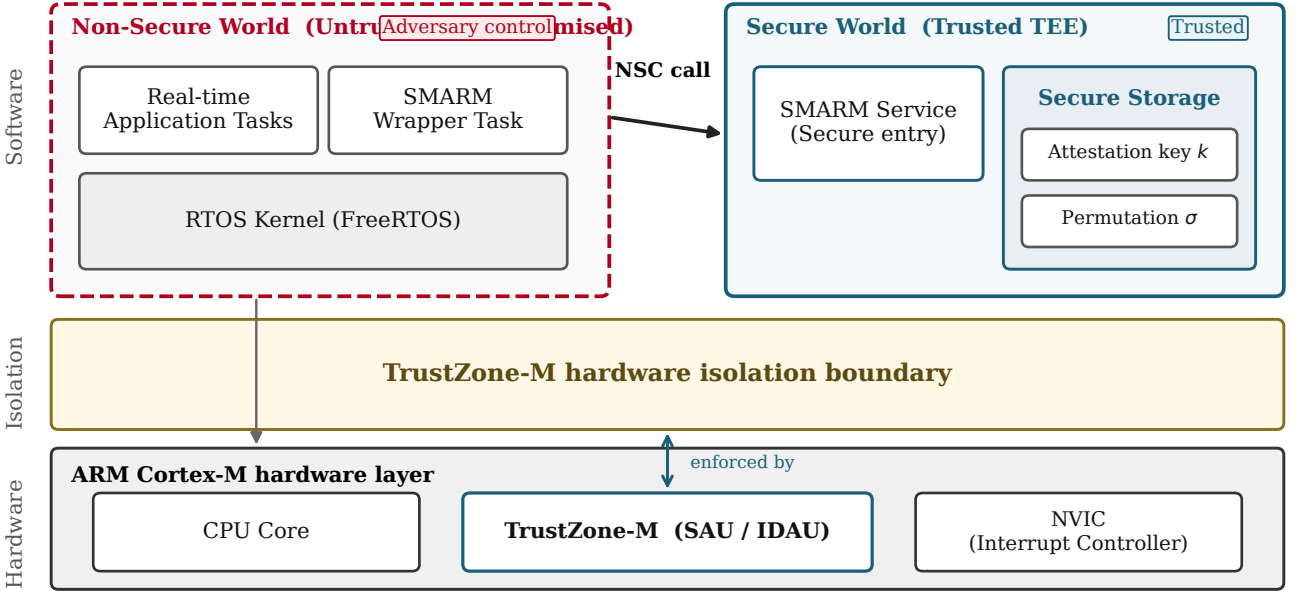
Software
Isolation
Hardware
Non-Secure World (Untrusted)
Adversary control
Real-time Application Tasks
SMARM Wrapper Task
RTOS Kernel (FreeRTOS)
NSC call
Secure World (Trusted TEE)
Trusted
SMARM Service (Secure entry)
Secure Storage
Attestation key k
Permutation σ
TrustZone-M hardware isolation boundary
enforced by
ARM Cortex-M hardware layer
CPU Core
TrustZone-M (SAU / IDAU)
NVIC (Interrupt Controller)